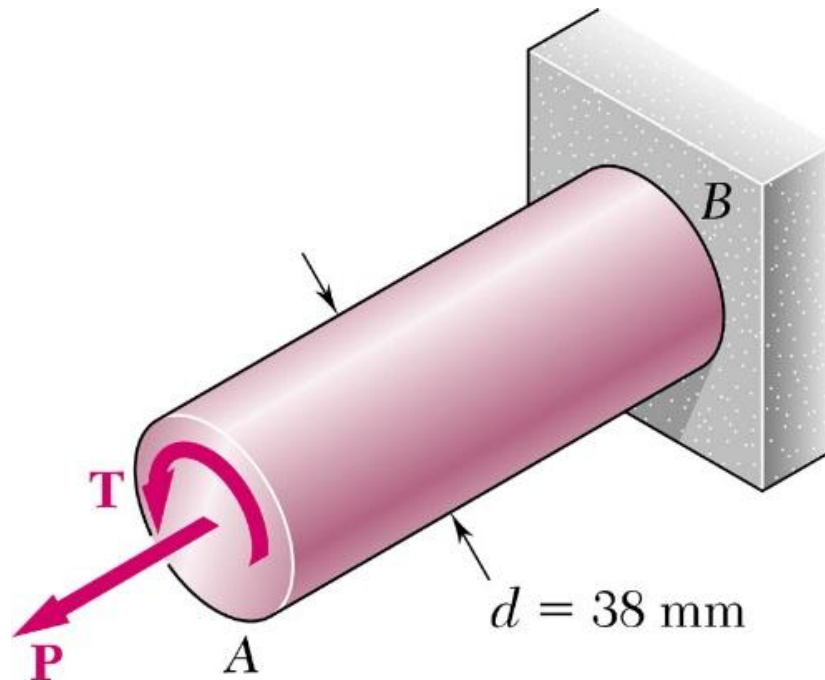


1. (Based on Beer and Johnston p7.87) The 38 mm diameter shaft AB at right is made of a grade of steel for which the yield strength is $\sigma_Y = 250 \text{ MPa}$. The applied axial load is $P = 120 \text{ kN}$. (a) Using the maximum-shearing-stress criterion determine the magnitude of the torque T for which yield occurs. (b) Estimate the allowable torque, T_{all} , if the desired factor of safety is 2.0. (c) Now estimate the torque at yield and the allowable torque using the maximum-distortion-energy criterion. Discuss your results (Compare the two criteria estimates. Which is more accurate?)



Given:

- $\sigma_Y = 250 \text{ MPa}$
- $P = 120 \text{ kN}$
- $d = 38 \text{ mm}$

Find:

- (a) Magnitude of torque T when yield occurs under the maximum shearing stress (TRESCA) criterion
- (b) Allowable torque T_{all} for a factor of safety of 2.0
- (a) and (b) using maximum-distortion energy (VON-MISES) criterion

Approach:

- Determine geometry:
 - Polar moment of inertia

$$J = \frac{\pi}{2} c^4, c = \frac{d}{2}$$

$$A = \frac{\pi}{4} d^2$$

- Determine state of stress at surface:
 - Shear stress from torque:

$$\tau_{z\theta} = \frac{Tc}{J}$$

- Axial stress from applied load:

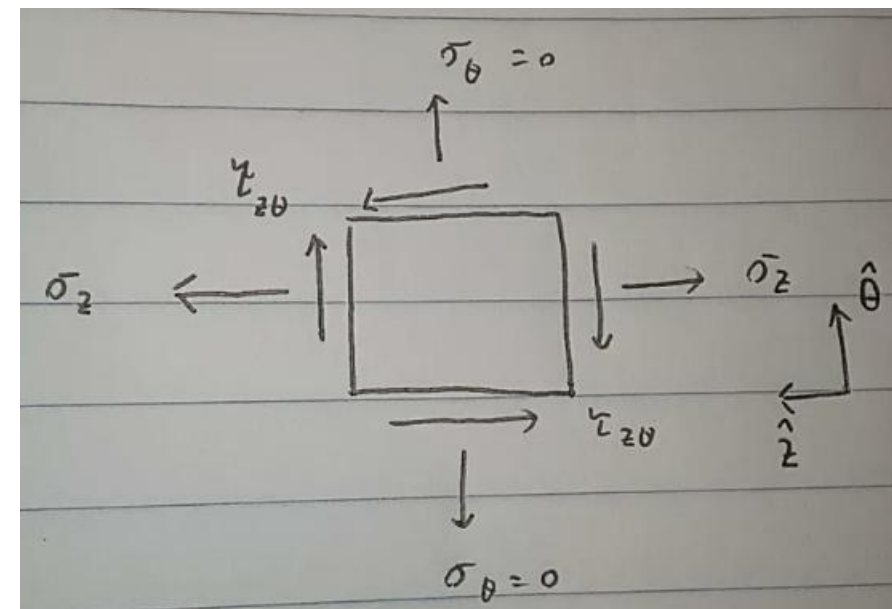
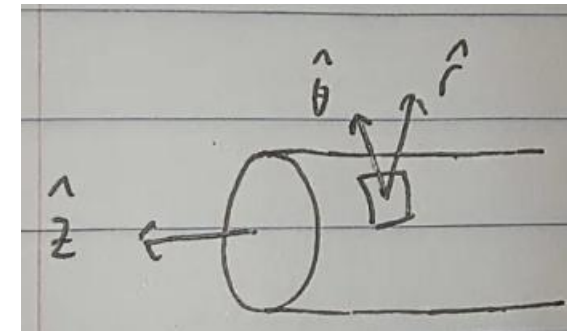
$$\sigma_z = \frac{P}{A}$$

- Determine principal stresses and max shear stress:

- Where $\Delta\sigma = \sigma_z - \sigma_\theta$, $\sigma_{ave} = (\sigma_z + \sigma_\theta)/2$:

$$R = \sqrt{\tau_{z\theta}^2 + \left(\frac{\Delta\sigma}{2}\right)^2}$$

$$\sigma_{1,2} = \sigma_{ave} \pm R, \quad \tau_{max} = R$$



Approach:

- Tresca Criterion

$$\tau_{max} \leq \frac{1}{2} \sigma_Y$$

- Rearrange to find $T_{max,T}$ for $\tau_{max} = \frac{1}{2} \sigma_Y$

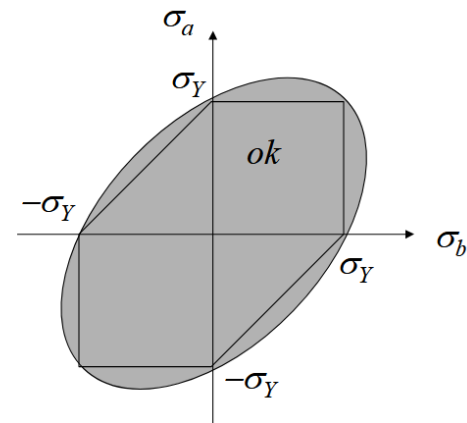
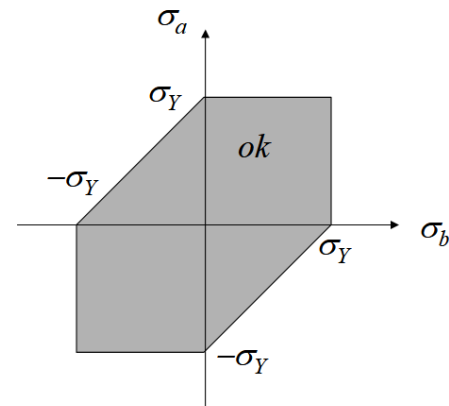
- Von-Mises Criterion

$$\sigma_1^2 - \sigma_1 \sigma_2 + \sigma_2^2 \leq \sigma_Y^2$$

- Rearrange to find $T_{max,VM}$ for $\sigma_1^2 - \sigma_1 \sigma_2 + \sigma_2^2 = \sigma_Y^2$

- Factor of Safety

$$FS = \frac{\sigma_Y}{\sigma_{all}} = 2$$



- Geometry:
 - Polar moment of inertia

$$J = \frac{\pi}{32} d^4 = \frac{\pi}{32} [0.038 \text{ m}]^4 = 2.047 \times 10^{-7} \text{ m}^4$$
$$A = \frac{\pi}{4} d^2 = \frac{\pi}{4} [0.038 \text{ m}]^2 = 1.134 \times 10^{-3} \text{ m}^2$$

- Determine state of stress at surface:
 - Shear stress from torque:

$$\tau_{z\theta} = \frac{Td}{2J}$$

- Axial stress from applied load:

$$\sigma_z = \frac{P}{A} = \frac{[120 \times 10^3 \text{ N}]}{[1.134 \times 10^{-3} \text{ m}^2]} = 105.82 \times 10^6 \text{ N/m}^2$$

- Determine principal stresses and max shear stress:
 - $\Delta\sigma = \sigma_z$, $\sigma_{ave} = (\sigma_z)/2$:

$$R = \sqrt{\tau_{z\theta}^2 + \left(\frac{\Delta\sigma}{2}\right)^2} = \sqrt{\left(\frac{Td}{2J}\right)^2 + \left(\frac{\sigma_z}{2}\right)^2}$$
$$\sigma_{1,2} = \sigma_{ave} \pm R, \quad \tau_{max} = R$$

Under the max shearing stress (Tresca) criterion

Find T_Y

$$\tau_{max} = \frac{1}{2} \sigma_Y$$

$$\sqrt{\left(\frac{T_Y d}{2J}\right)^2 + \left(\frac{\sigma_z}{2}\right)^2} = \frac{1}{2} \sigma_Y$$

$$T_Y = \frac{2J}{d} \sqrt{\frac{\sigma_Y^2 - \sigma_z^2}{4}}$$

$$T_Y = \frac{2[2.047 \times 10^{-7} m^4]}{[0.038 m]} \sqrt{\frac{\left[250 \times 10^6 \frac{N}{m^2}\right]^2 - \left[105.82 \times 10^6 \frac{N}{m^2}\right]^2}{4}} = 1220.118 Nm$$

$$\boxed{T_Y = 1220.118 Nm}$$

Find T_{all}

Based on allowable normal stress:

$$FS = \frac{\sigma_Y}{\sigma_{all}} = 2 \longrightarrow \sigma_{all} = \frac{\sigma_Y}{2} = \frac{1}{2} \left[250 \times 10^6 \frac{N}{m^2} \right] = 125 \times 10^6 \frac{N}{m^2}$$

$$\tau_{all} = \frac{1}{2} \sigma_{all} \longrightarrow T_{all} = \frac{2J}{d} \sqrt{\frac{\sigma_{all}^2 - \sigma_z^2}{4}} = \frac{2[2.047 \times 10^{-7} m^4]}{[0.038 m]} \sqrt{\frac{\left[125 \times 10^6 \frac{N}{m^2}\right]^2 - \left[105.82 \times 10^6 \frac{N}{m^2}\right]^2}{4}} = 358.42 Nm$$

$$\boxed{T_{all} = 358.42 Nm}$$

Under the maximum-distortion energy (von-Mises) criterion

Find T_Y

$$\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2 = \sigma_Y^2$$

$$(\sigma_1 + \sigma_2)^2 - 3\sigma_1\sigma_2 = \sigma_Y^2$$

$$((\sigma_{ave} + R) + (\sigma_{ave} - R))^2 - 3(\sigma_{ave} + R)(\sigma_{ave} - R) = \sigma_Y^2$$

$$(2\sigma_{ave})^2 - 3(\sigma_{ave}^2 - R^2) = \sigma_Y^2$$

$$\sigma_{ave}^2 + 3R^2 = \sigma_Y^2$$

$$\left(\frac{\sigma_z}{2}\right)^2 + 3\left[\left(\frac{T_Y d}{2J}\right)^2 + \left(\frac{\sigma_z}{2}\right)^2\right] = \sigma_Y^2$$

$$3\left(\frac{T_Y d}{2J}\right)^2 + 4\left(\frac{\sigma_z}{2}\right)^2 = \sigma_Y^2$$

$$T_Y = \frac{2J}{d} \sqrt{\frac{\sigma_Y^2 - \sigma_z^2}{3}} = \frac{2[2.047 \times 10^{-7} m^4]}{[0.038 m]} \sqrt{\frac{\left[250 \times 10^6 \frac{N}{m^2}\right]^2 - \left[105.82 \times 10^6 \frac{N}{m^2}\right]^2}{3}} = 1408.871 Nm$$

$$\boxed{T_Y = 1408.871 Nm}$$

Under the maximum-distortion energy (von-Mises) criterion

Find T_{all}

$$FS = \frac{\sigma_Y}{\sigma_{all}} = 2 \quad \longrightarrow \quad \sigma_{all} = \frac{\sigma_Y}{2} = \frac{1}{2} \left[250 \times 10^6 \frac{N}{m^2} \right] = 125 \times 10^6 \frac{N}{m^2}$$

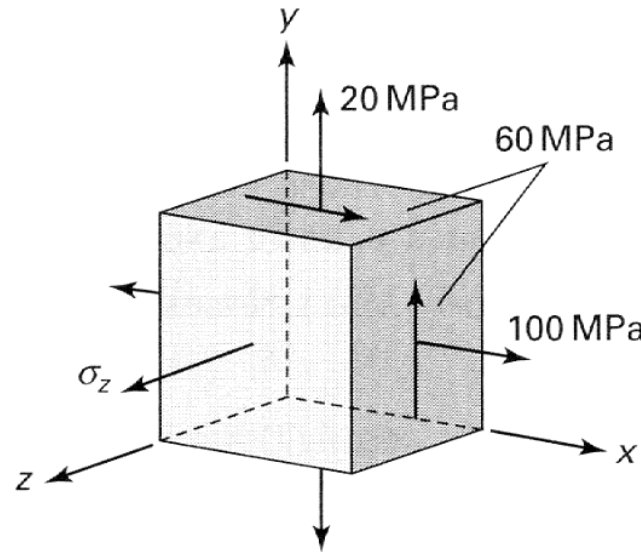
$$T_{all} = \frac{2J}{d} \sqrt{\frac{\sigma_{all}^2 - \sigma_z^2}{3}} = \frac{2[2.047 \times 10^{-7} m^4]}{[0.038 m]} \sqrt{\frac{\left[125 \times 10^6 \frac{N}{m^2} \right]^2 - \left[105.82 \times 10^6 \frac{N}{m^2} \right]^2}{3}} = 413.87 Nm$$

$$\boxed{T_{all} = 413.87 Nm}$$

The max shearing stress (Tresca) criterion provides a smaller allowable torque compared to the max distortion energy (von-Mises) criterion. This is because the range of safe stress values for the Tresca criterion overlaps and is contained entirely within the range for the von-Mises criterion. However, von-Mises is more accurate for an isotropic material with a uniaxial yield stress such as for metals. [Time spent: 45 mins]

2. (Based on Ugaral and Fenster 5th edition, p1.81) A point in a structural member is under 3D stress, with $\sigma_x = 100$ MPa, $\sigma_y = 20$ MPa, $\tau_{xy} = 60$ MPa, and $\sigma_z =$ (a) 30 MPa, or (b) -30 MPa.

- Guess based on principles, prior to calculation, which sign for σ_z will give the largest maximum shear stress.
- Write a MATLAB or Python computer routine to solve for principal stresses, maximum shear stresses, and orientation of principal planes for a 3D state of stress. (Check if your routine works by running a textbook example or using a Mohr's circle.) Also calculate the von Mises (effective) stress (see text section 4.6, un-numbered equation after eq (4.5b)).
- Use your routine to calculate the absolute maximum shear stress and von Mises stress, and sketch Mohr's circles, in both cases.
- Discuss how your maximum shear stress case guess in (i) worked, and if you learned anything from guessing.
- Also discuss the trend in the von Mises stress.



Approach to part i:

- From the 3d Mohr's circle, the maximum shear stress is given by

$$\tau_{abs,max} = \frac{\sigma_1 - \sigma_3}{2} = R_{max}$$

- By definition, a principal axis exists when there is a face normal to it that carries no shear stress (the $\hat{z}$ face), yielding a principal stress $\sigma_a = \sigma_z$
- The other principal stresses can be by rotating the stress element about the z-axis until the remaining shear stresses go to zero. The principal angle can be found using the plane stress formulas:

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\Delta\sigma}, \quad \Delta\sigma = \sigma_x - \sigma_y, \quad \sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$

- Finally, using θ_p , find the other two principal stresses (σ_b, σ_c) via stress transformation (given in terms of xy):

$$\begin{aligned} \sigma_b &= \sigma_{x'}(\theta_p) = \sigma_{ave} + \frac{\Delta\sigma}{2} \cos 2\theta_p + \tau_{xy} \sin 2\theta_p \\ \sigma_c &= \sigma_{y'}(\theta_p) = \sigma_{ave} - \frac{\Delta\sigma}{2} \cos 2\theta_p - \tau_{xy} \sin 2\theta_p \end{aligned}$$

Prior to calculation, these formulas tells us that the max shear stress results when the difference between the smallest and largest principal stresses is maximized. Since $\sigma_{ave} = 60 \text{ MPa}$, the center of the Mohr's circle for rotation around the z-axis will be tensile. Additionally, the radius for this Mohr's circle is $R = \sqrt{\tau_{xy}^2 + \left(\frac{\Delta\sigma}{2}\right)^2} = 72.11 \text{ MPa}$. Thus, the σ_b and σ_c principal stresses are tensile (larger than $|\sigma_z|$) and compressive ($-|\sigma_z| < [-12.11 \text{ MPa}]$) since $(\sigma_{b,c} = \sigma_{ave} \pm R)$. To further maximize the radius of the *3D Mohr's circle*, the third principal stress $\sigma_a = \sigma_z$ must be compressive with a sign of -30 MPa .

Approach to part ii:

- Define the stress tensor:

$$[\tau] = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

- Set up eigenvalue problem:

$$([\tau] - \sigma_p[I]) \cdot \hat{n} = \vec{0}$$

- Solving the eigenvalue problem gives us
 - σ_p : principal stresses (eigenvalues)
 - $\hat{n}$: principal axis direction for each principal stress (eigenvectors)
- Verification:
 - Construct the cosine matrix to reorient the stress element to be aligned with the principal axis directions

$$[L] = \begin{bmatrix} \hat{n}_1^{Tr} \\ \hat{n}_2^{Tr} \\ \hat{n}_3^{Tr} \end{bmatrix} = \begin{bmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{bmatrix}$$

- Plug in:

$$[\tau'] = [L][\tau][L]^{tr}$$

This should result in a matrix with only the 3 principal stresses along its diagonal as all shear stress goes to zero.

Approach to part ii:

- Find max shear stress

- Order the 3 principal stresses such that $\sigma_1 \geq \sigma_2 \geq \sigma_3$

$$\tau_{abs,max} = \frac{\sigma_1 - \sigma_3}{2} = R_{max}$$

- von Mises (effective) stress from eqn 4.4b. Where σ_e is the von Mises or effective stress:

$$2\sigma_e^2 = (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2$$

Textbook EX 1.6

A steel shaft is to be force fitted into a fixed-ended cast-iron hub. The shaft is subjected to a bending moment M , a torque T , and a vertical force P , Fig. 1.21a. Suppose that at a point Q in the hub, the stress field is as shown in Fig. 1.21b, represented by the matrix

$$\begin{bmatrix} -19 & -4.7 & 6.45 \\ -4.7 & 4.6 & 11.8 \\ 6.45 & 11.8 & -8.3 \end{bmatrix} \text{MPa}$$

Find: The principal stresses and their orientation with respect to the original coordinate system.

Solution

Substituting the given stresses into Eq. (1.33) we obtain from Eqs. (B.2)

$$\sigma_1 = 11.618 \text{ MPa}, \quad \sigma_2 = -9.001 \text{ MPa}, \quad \sigma_3 = -25.316 \text{ MPa}$$

Successive introduction of these values into Eq. (1.31), together with Eq. (1.30a), or application of Eqs. (B.6) yields the direction cosines that define the orientation of the planes on which σ_1 , σ_2 , and σ_3 act:

$$l_1 = 0.0266, \quad l_2 = -0.6209, \quad l_3 = 0.7834$$

$$m_1 = -0.8638, \quad m_2 = 0.3802, \quad m_3 = 0.3306$$

$$n_1 = -0.5031, \quad n_2 = -0.6855, \quad n_3 = -0.5262$$

Matlab output:

```
principal_stresses =
```

```
11.6178
-9.0015
-25.3163
```

```
principal_directions =
```

```
-0.0266    0.6209   -0.7834
 0.8638   -0.3802   -0.3306
 0.5031    0.6855    0.5262
```

```
tau_max =
```

```
18.4670
```

```
sigma_e =
```

```
32.0582
```

Matlab Code:

```
% Stress tensor
stress_tensor = [-19, -4.7, 6.45;
                 -4.7, 4.6, 11.8;
                 6.45, 11.8, -8.3];

% Solve eigenvalue problem
[V, D] = eig(stress_tensor);

% Principal stresses and directions
principal_stresses = diag(D);

% Sort principal stresses from largest to smallest
[principal_stresses, order] = sort(principal_stresses, 'descend');
principal_directions = V(:, order);

% Individual principal stresses (after sorting, 1 is the largest, 3 is the smallest)
sigma1 = principal_stresses(1);
sigma2 = principal_stresses(2);
sigma3 = principal_stresses(3);
```

```
% Absolute maximum shear stress
tau_max = (sigma1 - sigma3) / 2;

% Von Mises (equivalent) stress
sigma_e = sqrt(0.5 * ((sigma1 - sigma2)^2 + ...
                    (sigma2 - sigma3)^2 + ...
                    (sigma3 - sigma1)^2));

% Display results
principal_stresses
principal_directions
tau_max
sigma_e
```

Part iii: $\sigma_z = +30 \text{ MPa}$

```
stress_tensor =
```

```
100    60    0
 60    20    0
 0     0    30
```

```
principal_stresses =
```

```
132.1110
 30.0000
-12.1110
```

```
principal_directions =
```

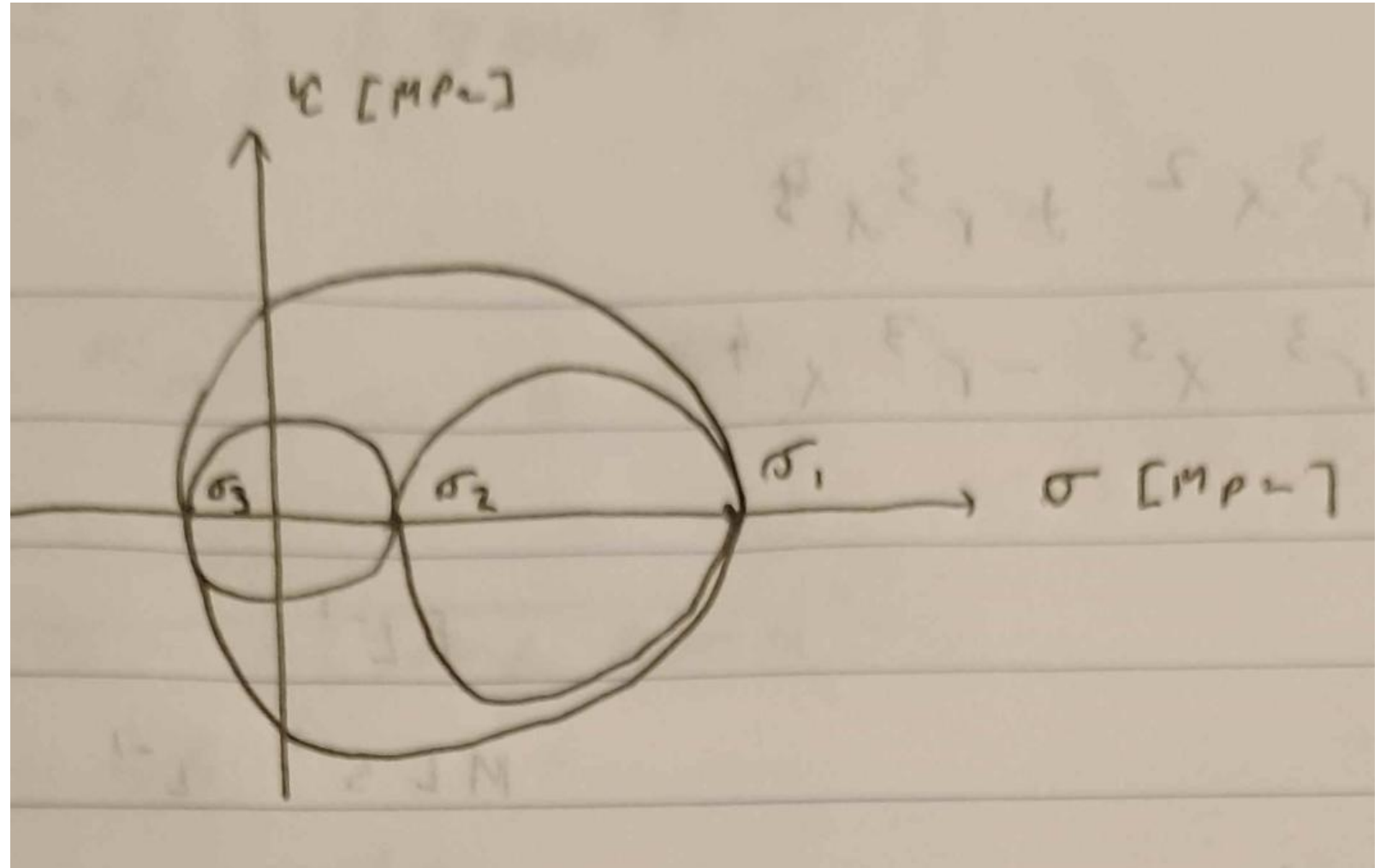
```
-0.8817    0    0.4719
-0.4719    0   -0.8817
 0     1.0000    0
```

```
tau_max =
```

```
72.1110
```

```
sigma_e =
```

```
128.4523
```



Part iii: $\sigma_z = -30 \text{ MPa}$

```
stress_tensor =
```

```
100    60    0
 60    20    0
  0     0   -30
```

```
principal_stresses =
```

```
132.1110
-12.1110
-30.0000
```

```
principal_directions =
```

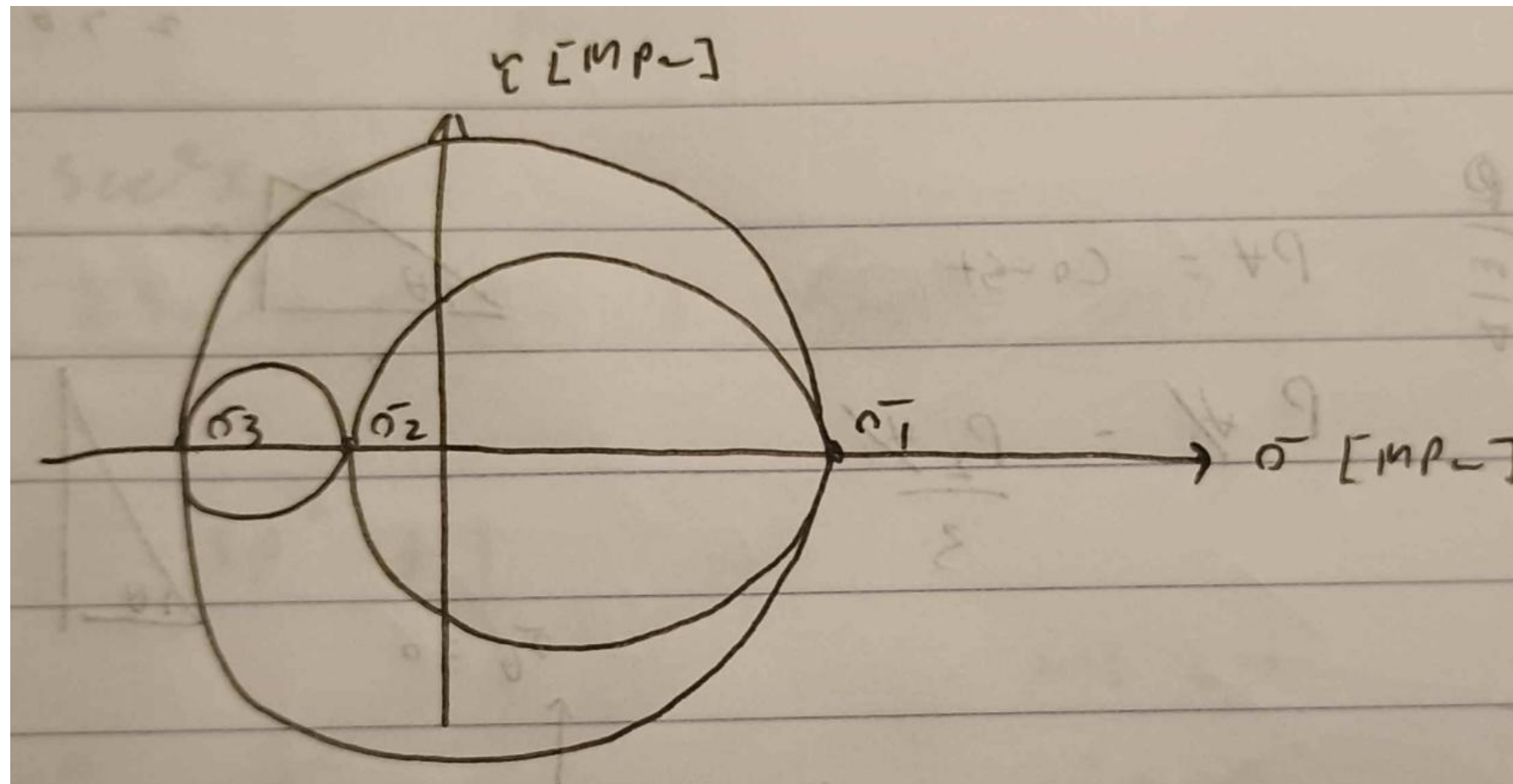
```
-0.8817  0.4719  0
-0.4719 -0.8817  0
  0      0      1.0000
```

```
tau_max =
```

```
81.0555
```

```
sigma_e =
```

```
153.9480
```



Zeshui Song, 9/17/26, ME 300 HW2

Iv: the guess from part I was correct, it expanded the lower bound of the 3D Mohr's circle and thus increased the max radius and max shear stress.

V: The von mises stress also increased as a result of the increase in the max shear stress from the changing of σ_z from tensile to compressive stress. This makes sense since for a tensile loading, the spread of the principal stresses are smaller than for a compressive loading as seen by the stress differences of : $\sigma_1 - \sigma_2$, $\sigma_2 - \sigma_3$, $\sigma_3 - \sigma_1$ in the formula.

[Time spent: 1 hr]

3. (Based on U&F, p 1.79) At a point in a loaded body, the stresses relative to an (x,y,z) coordinate system are

$$[\tau_{ij}] = \begin{bmatrix} 40 & 40 & 30 \\ 40 & 20 & 0 \\ 30 & 0 & 20 \end{bmatrix} MPa$$

Determine the normal stress and shear stress magnitudes on a plane whose outward normal is oriented at angles of 40° , 75° , and 54° relative to the x , y , and z axes, respectively. (This should be a simple calculation, especially using software.) (Hint: think about how you can use the traction vector and some simple vector operations to calculate these stress magnitudes.)

Approach:

- Define the direction cosine vector (defines the normal vector direction to the cut face)

$$\hat{n} = \begin{bmatrix} l \\ m \\ n \end{bmatrix} = \begin{bmatrix} \cos\alpha \\ \cos\beta \\ \cos\gamma \end{bmatrix}$$

- Define the traction vector using the direction cosine vector:

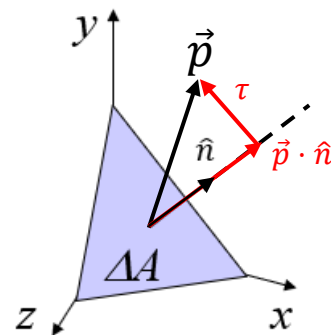
$$\vec{p} = [\tau] \cdot \hat{n}$$

- Find normal stress magnitude by projecting the traction vector onto the normal vector:

$$\sigma = |\vec{p} \cdot \hat{n}|$$

- Find shear stress magnitude by using the Pythagorean theorem (normal stress and shear stress are perpendicular):

$$\tau^2 + \sigma^2 = \|\vec{p}\|^2$$




```
% Stress tensor [MPa]
stress_tensor = [40, 40, 30;
                 40, 20, 0;
                 30, 0, 20];

% Direction cosines (unit normal vector)
n = [cosd(40);
     cosd(75);
     cosd(54)];

% Traction vector
p = stress_tensor * n;

% Normal stress magnitude [MPa]
sigma_mag = abs(dot(p, n));

% Shear stress magnitude [MPa]
tau_mag = sqrt(norm(p)^2 - sigma_mag^2);

% Display results
fprintf('Normal stress magnitude: %.2f MPa\n', sigma_mag);
fprintf('Shear stress magnitude: %.2f MPa\n', tau_mag);
```

```
Normal stress magnitude: 74.60 MPa
Shear stress magnitude: 19.02 MPa
```

```
>> eig(stress_tensor)

ans =

   -20.9902
    20.0000
    80.9902
```

This result makes sense because it does not exceed the principal stresses given by the stress tensor. Additionally, the normal stress is pretty close to one of the principal stresses, suggesting that the normal direction is close to a principal axis. [Time spent: 20 mins]

4. A short, thick wood beam is loaded in shear, bending, and torsion, and experiences clamping forces at the fixed end, leading to vertical and lateral compression stresses. At a critical location near the clamped end, the state of stress relative to the loading axes (see figure) is:

$$[\sigma_{ij}] = \begin{bmatrix} 3.5 & 1.0 & 1.2 \\ 1.0 & -1.5 & 0 \\ 1.2 & 0 & -1 \end{bmatrix} MPa$$

The direction of the wood grain is variable and not aligned with the structural axes. The stress normal to the grain, and the shear stresses on the plane of the grain parallel to and transverse to the wood fibers, are needed to estimate the risk of failure and factor of safety. At a critical location, a local orthogonal coordinate system (f, n, t) oriented with the wood grain fiber direction (unit vector $\hat{e}_f$), normal to the harder, darker, growth rings (unit vector $\hat{e}_n$), and transverse to the grain fibers and parallel to the growth rings (unit vector $\hat{e}_t$), respectively, has been established. A direction cosine matrix relating the beam coordinates with the local coordinate system has been measured for you and is

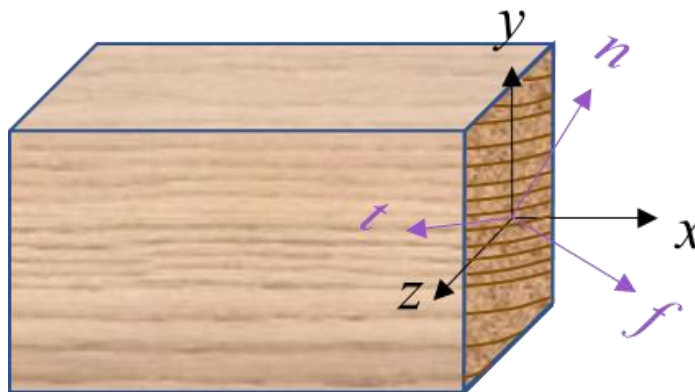
$$[A_{ij}] = \begin{bmatrix} \hat{e}_f \cdot \hat{e}_x & \hat{e}_n \cdot \hat{e}_x & \hat{e}_t \cdot \hat{e}_x \\ \hat{e}_f \cdot \hat{e}_y & \hat{e}_n \cdot \hat{e}_y & \hat{e}_t \cdot \hat{e}_y \\ \hat{e}_f \cdot \hat{e}_z & \hat{e}_n \cdot \hat{e}_z & \hat{e}_t \cdot \hat{e}_z \end{bmatrix} = \begin{bmatrix} 0.8529 & 0.4924 & -0.1736 \\ -0.4742 & 0.8697 & 0.1371 \\ 0.2185 & -0.0345 & 0.9752 \end{bmatrix}$$

Find the normal stress in the direction normal to the wood grain, the normal stress in the direction of the wood fibers, and the shear stresses parallel (nf in the figure), transverse (nt), and normal (tf) to the wood fiber direction.

(Note: check if the A matrix is the same as or different from the definition of the L matrix given in table 1.2 of the text.)

Approach:

This problem asks us to transform the stress tensor from a baseline xyz to a transformed fnt coordinate frame:



- Compare the given direction cosine matrix to textbook:

$$[A_{ij}] = \begin{bmatrix} \hat{e}_f \cdot \hat{e}_x & \hat{e}_n \cdot \hat{e}_x & \hat{e}_t \cdot \hat{e}_x \\ \hat{e}_f \cdot \hat{e}_y & \hat{e}_n \cdot \hat{e}_y & \hat{e}_t \cdot \hat{e}_y \\ \hat{e}_f \cdot \hat{e}_z & \hat{e}_n \cdot \hat{e}_z & \hat{e}_t \cdot \hat{e}_z \end{bmatrix}, \quad [L] = \begin{bmatrix} \hat{i}' \cdot \hat{i} & \hat{i}' \cdot \hat{j} & \hat{i}' \cdot \hat{k} \\ \hat{j}' \cdot \hat{i} & \hat{j}' \cdot \hat{j} & \hat{j}' \cdot \hat{k} \\ \hat{k}' \cdot \hat{i} & \hat{k}' \cdot \hat{j} & \hat{k}' \cdot \hat{k} \end{bmatrix}$$

The columns of [A] are the rows of [L]:

$$[A]^{Tr} = [L]$$

- Compute the transformed stress tensor:

$$[\tau'] = [L][\tau][L]^{tr}$$

In the notation of this problem:

$$[\sigma_{(f,n,t)}] = [A]^{Tr}[\sigma_{(x,y,z)}][A]$$

$$\sigma_{(f,n,t)} = \begin{bmatrix} \sigma_f & \tau_{nf} & \tau_{tf} \\ \tau_{fn} & \sigma_n & \tau_{tn} \\ \tau_{ft} & \tau_{nt} & \sigma_t \end{bmatrix}$$

```

% Stress tensor in baseline coordinates (x, y, z) [MPa]
sigma_xyz = [ 3.5,  1.0,  1.2;
              1.0, -1.5,  0.0;
              1.2,  0.0, -1.0];

% Given direction cosine matrix [A]
A = [ 0.8529,  0.4924, -0.1736;
      -0.4742,  0.8697,  0.1371;
       0.2185, -0.0345,  0.9752];

% Textbook direction cosine matrix [L] = [A]'
L = A';

% Transformed stress tensor in local coordinates (f, n, t)
sigma_fnt = L * sigma_xyz * L';

sigma_f = sigma_fnt(1, 1); % Wood fiber direction
sigma_n = sigma_fnt(2, 2); % Normal to wood grain
tau_nf  = sigma_fnt(2, 1); % Parallel to fibers on grain plane
tau_nt  = sigma_fnt(2, 3); % Transverse to fibers on grain plane
tau_tf  = sigma_fnt(3, 1); % Normal to fiber direction

% Display results
fprintf('Normal stress in fiber direction (sigma_f):      %7.3f MPa\n', sigma_f);
fprintf('Normal stress normal to grain (sigma_n):          %7.3f MPa\n', sigma_n);
fprintf('Shear stress parallel to fibers (tau_nf):              %7.3f MPa\n', tau_nf);
fprintf('Shear stress transverse to fibers (tau_nt):            %7.3f MPa\n', tau_nt);
fprintf('Shear stress normal to fiber direction (tau_tf):       %7.3f MPa\n', tau_tf);

```

```

Normal stress in fiber direction (sigma_f):      1.799 MPa
Normal stress normal to grain (sigma_n):         0.529 MPa
Shear stress parallel to fibers (tau_nf):         2.698 MPa
Shear stress transverse to fibers (tau_nt):       0.056 MPa
Shear stress normal to fiber direction (tau_tf):  0.518 MPa

```

Since the highest stress is shear parallel to the wood fibers, this loading is relatively safe as wood is strong along the grain in the f direction but is weak against the grain in the n and t directions.
[Time spent: 20 mins]